

Final Examination, Business 33801:

(Monday, 15 January 2024)

Read all questions carefully and write answers in the space provided. If you run out of space, you may write on the back of the exam pages; if you run out of space on the back of the exam, you may add additional pages of paper. Graphs must be thoroughly labeled for full credit. Show your work and explain all equations on numeric questions. Be concise – irrelevant remarks will not count for anything. Be careful to budget your time between various parts, taking into account the points allocated to each question (the number in parentheses); these points vary depending upon the question.

There are 14 questions with a total of 180 points possible. You have 180 minutes for the examination, so assign about one minute per point. Allocate your time wisely.

YOU MUST READ AND SIGN THE HONOR STATEMENT BELOW IN ORDER TO RECEIVE CREDIT FOR THE EXAMINATION.

I pledge my honor that I have not violated the Honor Code during this examination.

I also pledge my honor that I will not disclose any information about the examination to anyone outside of EXP30 until after 30 January 2024.

Sign your name here.

True-False-Uncertain (40 points total)

Indicate whether the italicized portion of each statement is True, False, or Uncertain and EXPLAIN your answer. Read each question carefully. You may take the non-italicized portion of the statement as true. **No credit will be given for answers without explanation.**

Question 1 (8 points) True, False, Uncertain? [Explain your answer.] One of the policies discussed in the U.S. health-care debates was a proposal for the government to provide every person an annual payment of \$2500 per year to be used toward medical care. Such a plan allows individuals to choose how they wish to allocate their health-care dollars among various goods and services. *If health care is a constant-cost industry then, in the long run, the subsidy will be fully passed along to the health-care industry through increased prices.*

Question 2 (8 points) True, False, Uncertain? [Explain your answer.] Ronald is given a choice between a “sure thing” that pays £1000 and a gamble that pays £3500 with probability $\phi_1 = \frac{1}{4}$ and £500 with probability $\phi_2 = \frac{3}{4}$. Ronald prefers the gamble to the sure thing. *Ronald is risk loving.*

Question 3 (8 points) True, False, Uncertain? [Explain your answer.] *In the long-run the economic incidence of taxation falls more on producers than consumers, because demand is more elastic in the long-run than in the short-run.*

Question 4 (8 points) True, False, Uncertain? [Explain your answer.] Suppose that a firm’s cost of capital equipment decreases, but this decrease does not change the firm’s short-run or long-run marginal cost functions. *Then neither the firm’s short-run nor its long-run supply curve changes.*

Question 5 (8 points) True, False, Uncertain? [Explain your answer.] The government imposes a unit tax on a profit-maximizing monopolist; the monopolist must legally pay the tax to the government. *The monopolist’s revenues collected from consumers (the new retail price the monopolist charges consumers times the quantity sold) will necessarily decrease because of the tax.*

Short questions (140 points)

Question 6 (10 points) Jorf Enterprises is a price-taking firm that makes T-shirts at an opportunity cost of $C^o(q) = 50 + 2q + q^2$. [Hint, $MC(q) = 2 + 2q$.] If T-shirts are sold in a competitive market at the price $p = 20$, how many T-shirts should Jorf Enterprises produce?

Question 7 (20 points) Cinnamon & Sparrow (CS) are a local maker and seller of designer handbags, and it has monopoly power. Suppose that there are two types of consumers in equal proportion who potentially buy Cinnamon & Sparrow’s handbags: high demand (H) and low demand (L). High-demand consumers value a handbag at $v_H = 1000$ while low-demand consumers have a value of only $v_L = 500$. The marginal and average opportunity costs of manufacturing a handbag are constant and equal to $MC = AC = 100$.

(a). If C&S must set a single price, what is its optimal price? Explain.

(b). In an effort to segment its customers and increase profits, C&S is considering scuffing some of its handbags to reduce their quality. It costs nothing to scuff a bag, but a scuffed bag is only worth 600 to a high-demand consumer, while it is worth 450 to a low-demand consumer. If C&S can sell the “premium”

(i.e., unscuffed) handbags for a different price than the scuffed bags, what are the profit-maximizing prices of the two types of handbag? Explain.

Question 8 (20 points) Consider a market with demand given by $Q^d = 100 - 2p$ (alternatively, $p = 50 - \frac{1}{2}Q^d$).

(a). Suppose that there are currently 100 price-taking firms who may produce in the market, each with the same costs given by $C^o(q) = 2q$ for $q \leq 1$ (i.e., each firm's production capacity is limited to 1 unit). Describe the short-run market supply curve. What is the short-run equilibrium price in this competitive market?

(b). Suppose instead that there are only 2 firms competing (A and B). The cost function of each firm is identical and given by $C^o(q) = 2q$, with unlimited capacity. What are the optimal outputs for each firm in the equilibrium of this Cournot output game?

Question 9 (16 points) Tractors are sold in a competitive, constant-cost industry with an unlimited number of potential firms, each with the long-run opportunity cost function $C_{lr}^o(q) = 100 + 4q^2$. [Hint: the long-run marginal cost is $8q$.]

(a). What is the long-run equilibrium price in this market? [You may assume that tractors are transacted in the long run market equilibrium.]

(b). If the government gives a \$10 per-unit subsidy to the firms in the market, what is the long-run equilibrium price that buyers will pay.

Question 10 (16 points) Consider the competitive market for nickel and suppose that the demand and supply functions are represented by

$$Q^d = 30 - p,$$

$$Q^s = 6 + 3p.$$

(a). What is the equilibrium price and quantity of nickel?

(b). What is the market elasticity of demand in equilibrium?

(c). If additional nickel mines are discovered causing the supply for nickel to shift outward, will market expenditures for nickel increase or decrease?

Question 11 (12 points) Consider a monopolist with demand curve given by $p = 50 - 2q$ and opportunity cost function $C^o(q) = 2q$.

(a). What is the monopolist's optimal price?

(b). How much deadweight loss is generated by the monopolist relative to the socially-efficient output? [Hint: draw a graph.]

Question 12 (10 points) Suppose that a dominant (strategic) firm with constant unit costs of $MC = AC = 20$ faces a competitive fringe of price taking firms whose aggregate supply is $S(p) = 5p$. If output is homogenous and market demand is characterized by $p = 240 - Q^d$, how much should the dominant firm supply?

Question 13 (24 points) Consider a monopoly railroad that has two classes of customers – farmers who ship grain and oil companies that ship oil. Each segment has the same demand curve for freight service. The farmer’s demand curve for freight is given by

$$p_f = 120 - q_f,$$

and the oil companies demand curve for freight is given by

$$p_o = 120 - q_o.$$

Suppose that the railroad monopoly can provide up to 100 units of freight service at a constant marginal cost of 10, but it cannot ship any freight above 100 units in total. [For simplicity, you may ignore the shutdown/operate rule for this entire question.]

(a). If the monopolist can charge different freight prices for grain and oil (i.e., practice direct price discrimination), what are the optimal prices and outputs for the railroad to choose for each customer segment?

(b). Suppose that the government has decided to regulate the railroad’s price for farm freight by the railroad to charge farmers $\bar{p}_f = 40$, and requiring that the railroad supply all of the farmers’ demands at this price before it can sell any rail service to the oil companies. There are no price regulations for the oil-freight segment. What is the railroad’s optimal price and quantity for its oil customers?

(c). Suppose (as in (b)) that the government has decided to regulate the railroad’s price for farmers by setting the price at $\bar{p}_f = 40$, but (unlike part (b)) the government cannot force the railroads to serve all of the farmers’ demands at this price. The only requirement on the railroad is that any output sold to the farmers must be sold at $p_f = 40$. What is the railroad’s optimal price and quantity for its oil customers?

[Hint: Determine the marginal revenue of the farmers’ segment under this regulation and think about how this must relate to the marginal revenue of the oil segment.]

Question 14 (12 points) Suppose that a monopoly firm sells to 2 groups of consumers. The demands of each segment are given by the following:

$$P_1(q_1) = 100 - 2q_1,$$

$$P_2(q_2) = 200 - 2q_2.$$

The marginal cost of production is the same for each segment and is constant, but market 2 requires an additional transportation expense of 20 per unit sold (i.e., $MC_2 = MC_1 + 20$). There are no other costs.

(a). What can be said about the difference between the marginal revenues in markets 1 and 2 at the profit-maximizing prices and quantities? Specifically, what is $MR_2(q_2^*) - MR_1(q_1^*)$?

(b). What is the difference in optimal prices at the profit-maximizing prices and quantities? Specifically, what is $p_2^* - p_1^*$? [Hint: first determine the difference in optimal outputs, $q_2^* - q_1^*$, and then use the demand equations to determine the difference in optimal prices.]

Suggested answers to examination

1 False. A constant-cost industry never absorbs a subsidy (or a tax, for that matter). Thus, health care prices would remain unchanged and the recipients of the subsidy would benefit from the full amount of \$2500 in health care. Aside: This subsidy is not a per-unit subsidy, so if the industry were not constant cost and prices increased from the increase in demand, then the analysis would be more complicated.

2 Uncertain. According to the facts, Ronald prefers a gamble with a higher expected payoff ($E[\tilde{x}] = 0.25(3500) + 0.75(500) = 1250 > 1000$.) This is consistent with Ronald being risk loving, or being risk averse (but not too risk averse).

3 False, uncertain. While demand is more elastic in the long run, supply is also typically more elastic in the long run, so it is not clear how the burden changes from the short-run to the long-run. Indeed, in a constant-cost industry, the long-run supply elasticity is infinite, so the burden is entirely borne by the consumers.

4 False. In the short-run, the supply curve does not change because the cost of capital is sunk and is not involved in calculating the firm's short-run supply curve. In the long-run, however, the firm's AC curve shifts downward, which implies that the firm's long-run supply curve starts at a lower price point (i.e., lower minimum AC). Hence, the long-run supply curve changes.

5 True. The imposition of the unit tax will raise the price the monopolist charges the consumer because it is equivalent to an increase in marginal cost. [Aside: Whether the increase in price is by more or less than the amount of the tax is not relevant.] Because a monopolist always prices in the elastic region of the demand curve, a higher price translates into less consumer revenue.

6 We first use the marginal output rule. Because the firm is competitive, $MR(q) = p = 20$. Setting this equal to marginal cost yields

$$20 = 2 + 2q.$$

Solving for q , we find that if the firm produces, it should produce at $q^* = 9$. We next need to check that the firm should indeed produce. At $q^* = 9$, revenue is $R = 9 \cdot 20 = 180$. Cost is $C^o(9) = 50 + 18 + 81 = 149$. Hence, the firm should operate because it earns positive economic profit at $q = 9$.

7 (a). There are two possible prices to consider: $p = 1000$ and $p = 500$. At a price of $p = 1000$, sales are only made to the high type and per-capita profit is $\frac{1}{2}(1000 - 100) = 450$. Alternatively, at a price of $p = 500$, handbags are sold to both types, and per capita profit is $\pi = 500 - 100 = 400$. Hence, $p^* = 1000$ is optimal and handbags are only sold to the high types. Expected per-consumer profit is 450.

(b). Here, I will answer the question using the nonlinear pricing approach discussed in class and the Dolan and Simon chapter. The marginal valuations of the consumers can be represented as

unit	High value consumer	Low value consumer
scuffed bag	600	450
premium bag upgrade	400	50

Now consider the optimal price for each "good". For the scuffed bag, the per-capita profit from charging 600 is $\frac{1}{2}(600 - 100) = 250$. The per-capita profit from charging 450 is $(450 - 100) = 350 > 250$. Hence, $p_1^* = 450$. For the second unit of quality, the price of 400 generates a per-capita profit of $\frac{1}{2}(400 - 0) = 200$ (note that there is no marginal cost for this unit), and the per-capita profit of charging 50 is only $50 - 0 = 50$. Hence, $p_2^* = 400$. The result is that the full price for the premium bag is $p_{premium}^* = 450 + 400 = 850$, and the price for the scuffed bag is $p_s^* = 450$. [Aside: Total per-capita profit from the scuffed-bag strategy is $350 + 200 = 550$ which is 100 higher than the result in (a).]

8 (a). Each firm will supply 1 unit for any price above $MC = 2$ and zero for any price below 2. Thus, the entire market will supply 100 units for any price above 2 and zero for any price below 2. Graphically, it is a vertical line at $Q = 0$ for any $p < 2$; it is a horizontal line at $p = 2$ up to 100; it is then a vertical line at $Q = 100$. The demand curve will intersect this supply curve at $p^* = 2$. (Note that for 100 to be demanded and for all firms to be at capacity, price would have to be $p = 0$, but then no firm would produce, so it cannot intersect on the vertical portion).

(b). Note that market demand can be denoted

$$p = 50 - \frac{Q^d}{2}.$$

Let $\hat{q}_b$ be firm A 's conjecture about what B will bring to market. Then firm A 's residual demand is

$$p = 50 - \frac{\hat{q}_b + q_a}{2}.$$

Simplifying, firm A 's residual demand is

$$p = \left(50 - \frac{\hat{q}_b}{2}\right) - \frac{q_a}{2}.$$

Firm A 's MR is thus

$$MR_a(q_a) = \left(50 - \frac{\hat{q}_b}{2}\right) - q_a.$$

Setting this equal to $MC = 2$, we have

$$\left(50 - \frac{\hat{q}_b}{2}\right) - q_a = 2,$$

or

$$q_a = 48 - \frac{\hat{q}_b}{2}.$$

Applying a similar analysis for firm B yields

$$q_b = 48 - \frac{\hat{q}_a}{2}.$$

A Nash equilibrium, (q_a^*, q_b^*) , therefore requires that

$$q_a^* = 48 - \frac{1}{2}q_b^*,$$

$$q_b^* = 48 - \frac{1}{2}q_a^*.$$

Thus, $q_a^* = q_b^* = 32$, and aggregate output is

$$Q^* = q_a^* + q_b^* = 64.$$

Inserting this into the demand function, we obtain

$$p^* = 50 - \frac{64}{2} = 16.$$

9 (a). Because this is a constant-cost industry, we know that the long run supply curve will be a horizontal line equal to minimum long-run average cost. To find long-run average cost, we firsts find the value of output such that average cost is at a minimum. The formula is $MC(\hat{q}) = AC'_{lr}(\hat{q})$. Thus,

$$8\hat{q} = \frac{100}{\hat{q}} + 4\hat{q}.$$

Solving for $\hat{q}$, we obtain $\hat{q} = 5$. At this value, $\min AC'_{lr} = 40$. Thus, $p^* = 40$.

(b). Because the slope of the supply curve is zero, all of the subsidy will be passed along to buyers. The sellers will still receive $p_s^* = 40$, but now the buyer's will only pay $p_b^* = 40 - 10 = 30$.

10 (a). Setting $Q^s = Q^d$ implies $30 - p = 6 + 3p$, so $24 = 4p$ or $p^* = 6$. At this price, $Q^* = 30 - 6 = 24$.

(b). $\varepsilon^d = -\frac{\Delta Q^d}{\Delta p} \frac{p^*}{Q^*} = (-1) \frac{6}{26} = -0.25$.

(c). Because demand is inelastic, additional increases in output due to increased supply (i.e., movements downward along the demand curve) will lead to decreased market expenditures.

11 (a). The monopolist sets marginal revenue equal to marginal cost. In this case, we have

$$MR(q) = 50 - 4q \text{ and } MC(q) = 2.$$

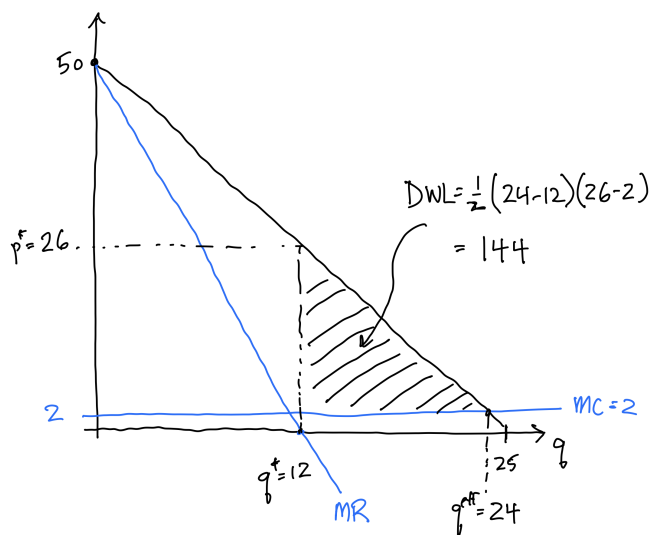
Thus, the marginal output rule requires

$$50 - 4q = 2,$$

or $q^* = 12$. At this output, the monopolist's optimal price is $p^* = 50 - 2(12) = 26$.

(b). [Note: you did not get full marks if you answered this question by using the trick from class that $CS = DWL$. As I told you in class, this trick is simply to check your work.]

The DWL is the area of the triangle between the demand curve and the marginal cost curve, from $q^* = 12$ to q^{eff} .



First note that q^{eff} is determined by the output for which MC intersects demand. Hence, $50 - 2q^{eff} = 2$, or $q^{eff} = 24$. Thus, the base of the triangle is $24 - 12 = 12$. The height of the triangle is $p^* - 2 = 26 - 2 = 24$. Thus, $DWL = \frac{1}{2}(12)(24) = 144$.

12 The dominant firm's residual demand curve is given by

$$p = 240 - (q + 5p).$$

Simplifying, we have $6p = 240 - q$, or

$$p = 40 - \frac{q}{6}.$$

Applying monopoly tools to this residual demand, the marginal output rule gives us

$$MR(q) = 40 - \frac{q}{3} = MC = 20,$$

so $\frac{q}{3} = 20$ or $q^* = 60$.

13 (a). Because the demand curves are identical, the marginal revenue curves are also identical. As a consequence, the monopolist will want to allocate equal the same amount of freight to both segments. There are two possibilities for the optimal allocation. If setting marginal revenue equal to 10 (i.e., marginal cost below capacity) in each segment will lead to output of less than 100, then this is optimal. Otherwise the monopolist should allocate $q_o^* = q_f^* = 50$ to each segment. Consider the farm segment first. The marginal output rule implies that

$$MR_f(q_f) = 120 - 2q_f = 10,$$

so $q_f = 55$. The same would be optimal for the oil segment. But producing 55 for each segment violates the railroad capacity constraint, so it will choose $q_f^* = q_o^* = 50$ instead.

(b). Because $\bar{p}_f = 40$ and the monopolist must satisfy the farmers' demands at this price, $q_f^* = 120 - 40 = 80$, the railroad is left with only 20 units of capacity. The marginal revenue of the oil segment at $q_o^* = 20$ is $MR_o(20) = 120 - 2(20) = 80$. Because this exceeds marginal cost, the railroad will want to allocate the remaining capacity to the oil segment. (You could have also referred to part (a) above and noted the railroad would clearly prefer to sell as much as 55 units to the oil segment, so it would certainly sell the remaining 20 units.) The optimal price, therefore, will be $p_o^* = 120 - 20 = 100$.

(c). In the farmers' market, the fixed-price regulation causes the railroad to earn a constant marginal revenue of 40 per unit, up to the farmers' demand at $\bar{p}_f = 40$ which is 80 units of freight. [Aside: Beyond 80 units of freight, the marginal revenue curve jumps downward to become negative. For example, the 81st unit earns a direct revenue contribution of price 39, but reduces the revenue on 80 previous units by 1, and thus marginal revenue decreases from +40 to -41 (i.e., a reduction of 81).]

Given the railroad always obtains a marginal revenue of 40 from the farming segment for any output $q_f < 80$, it will choose q_o such that the marginal revenue in the oil segment is equal to the marginal revenue in the farming segment:

$$MR_o(q_o) = 120 - 2(q_o) = 40.$$

Thus, $q_o^* = 40$. At this quantity, the corresponding optimal price for the oil segment is $p_o^* = 120 - 40 = 80$. An alternative approach is to note that the railroad's marginal opportunity cost of a unit of output sold to the oil segment is the revenue of 40 it could have earned on the farming segment. Thus, setting $MR_o = MC = 40$ yields the same answer.

[Aside: Because the railroad allocates 40 units of freight to the oil segment in part (d), only 60 units remain for the farming segment to be sold at $\bar{p}_f = 40$. At this regulated price, however, we know that farmers demand a total of 80 units, so demand will exceed supply.]

14 (a). The marginal output rule is applied to each segment. Thus, we must have

$$MR_1(q_1) = 100 - 4q_1 = MC$$

and

$$MR_2(q_2) = 200 - 4q_2 = MC + 20.$$

As such, $MR_2(q_2^*) = MR_1(q_1^*) + 20$ or $MR_2(q_2^*) - MR_1(q_1^*) = 20$.

(b). Solving the marginal output rules, note that we obtain two equations and two unknowns: $MC = 100 - 4q_1$ and $MC + 20 = 200 - 4q_2$. Substituting out MC , we obtain

$$(100 - 4q_1) + 20 = 200 - 4q_2,$$

or $q_2 - q_1 = 20$. Thus, $q_2^* = q_1^* + 20$. Substituting into the demand expressions, we have

$$p_1^* = 100 - 2q_1^*,$$

$$p_2^* = 200 - 2(q_1^* + 20).$$

Hence,

$$p_2^* - p_1^* = 200 - 2(q_1^* + 20) - (100 - 2q_1^*) = 200 - 40 - 100 = 60.$$